

FARMING ANIMALS

Java Program Documentation

1. Question

Write a Java program to create farm animal objects and display their name, place where they stay, food they eat and the sound they make.

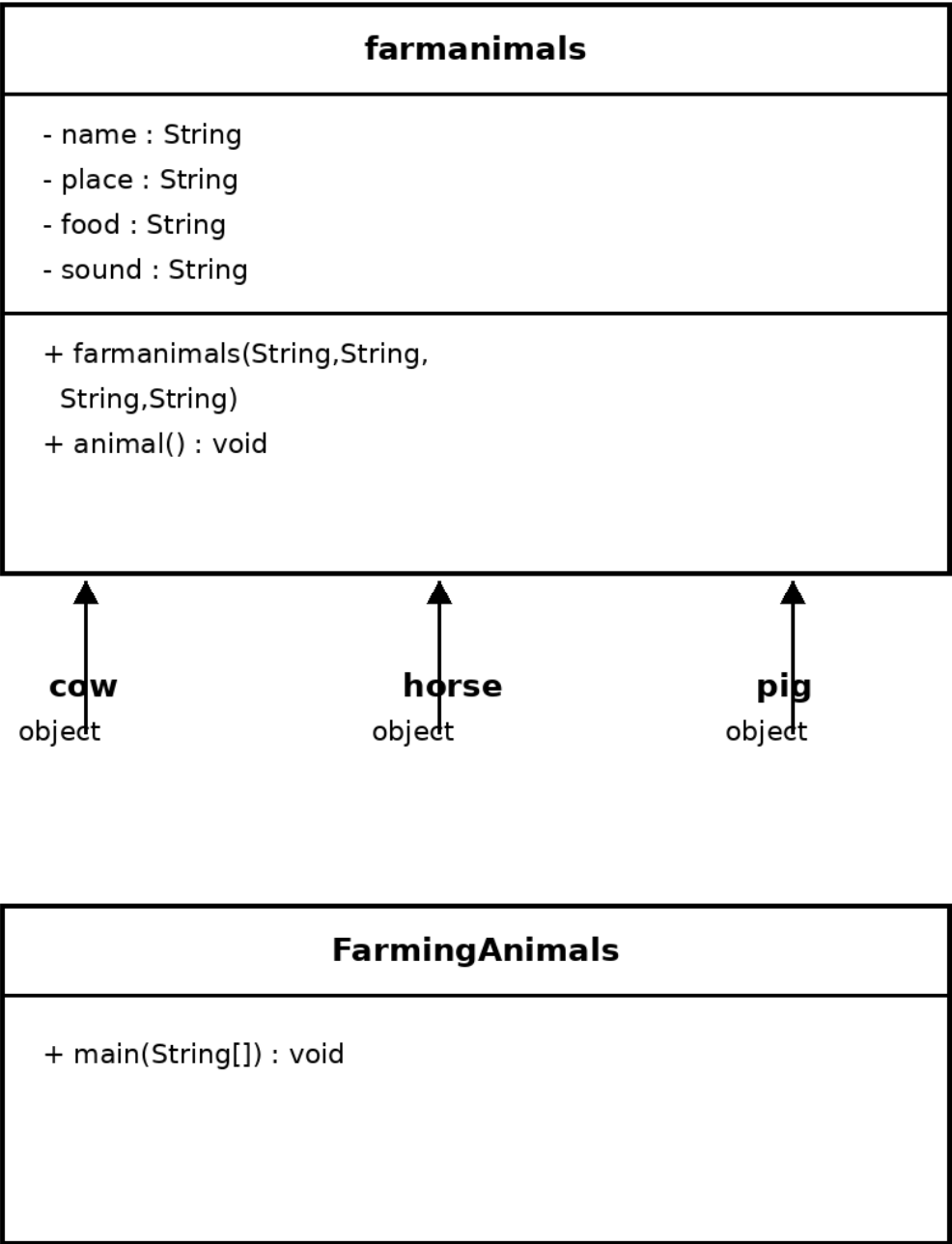
2. Execution Steps

1. Open Eclipse IDE or a terminal.
2. Create a Java project/package named Excercise.
3. Create the classes farmanimals and FarmingAnimals.
4. Copy the complete program into FarmingAnimals.java.
5. Make sure the public class name is FarmingAnimals.
6. Save the program.
7. Compile the program.
8. Run the FarmingAnimals class.
9. Three farm animal objects are created: cow, horse and pig.
10. Call animal() for each object and observe the displayed details.

3. Algorithm

1. Start.
2. Define the farmanimals class with name, place, food and sound variables.
3. Create a constructor to initialize the four values.
4. Define animal() to print the animal name, place, food and sound.
5. In main(), create a cow object with shed, grass and moo.
6. Create a horse object with stable, hay and neigh.
7. Create a pig object with pigsty, grains and oink.
8. Call animal() for the cow object.
9. Call animal() for the horse object.
10. Call animal() for the pig object.
11. Stop.

4. UML Class Diagram



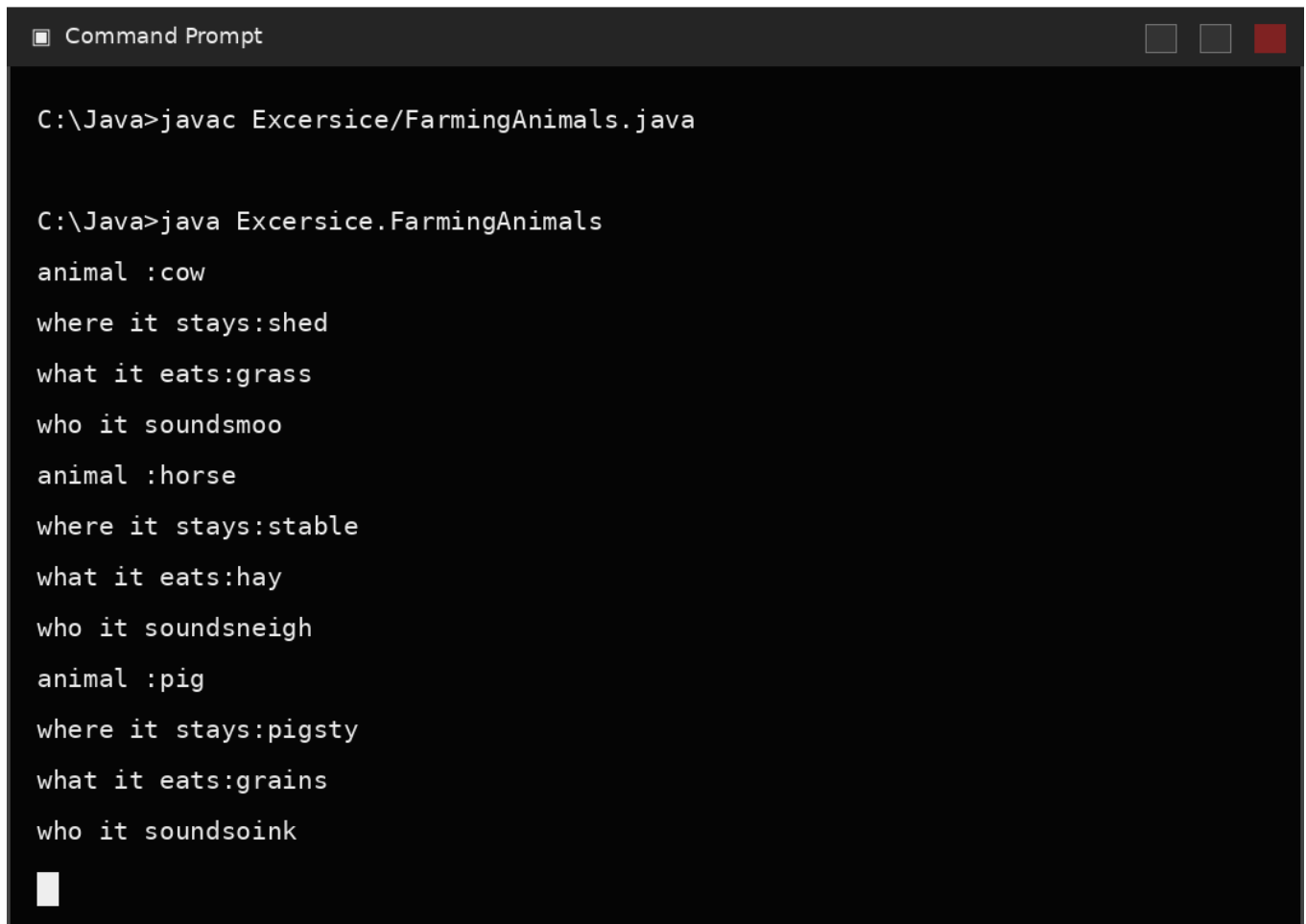
5. Java Source Code

```
package Excercise;

class farmanimals{
    String name;
    String place;
    String food;
    String sound;
    farmanimals(String k,String l,String m,String n){
        this.name=k;
        this.place=l;
        this.food=m;
        this.sound=n;
    }
    void animal() {
        System.out.println("animal :"+name);
        System.out.println("where it stays:"+place);
        System.out.println("what it eats:"+food);
        System.out.println("who it sounds"+sound);
    }
}

public class FarmingAnimals {
    public static void main(String[] args) {
        farmanimals cow = new farmanimals("cow","shed","grass","moo");
        farmanimals horse = new farmanimals("horse","stable","hay","neigh");
        farmanimals pig = new farmanimals("pig","pigsty","grains","oink");
        cow.animal();
        horse.animal();
        pig.animal();
    }
}
```

6. Output – Terminal Screenshot

A screenshot of a Windows Command Prompt window titled "Command Prompt". The window has a dark background and standard window controls (minimize, maximize, close) in the top right corner. The command prompt shows the following sequence of commands and output:

```
C:\Java>javac Excercise/FarmingAnimals.java

C:\Java>java Excercise.FarmingAnimals
animal :cow
where it stays:shed
what it eats:grass
who it soundsmoo
animal :horse
where it stays:stable
what it eats:hay
who it soundsneigh
animal :pig
where it stays:pigsty
what it eats:grains
who it soundsoink
```

The cursor is visible at the end of the last line of output.